

# ALTOS® HD Lite, Gel-Free, Single-Jacket, Single-Armored Cable with FastAccess® Technology 144F, SMF-28® Ultra fiber, Single-mode (OS2)

CORNING

**Part Number:**  
**144ZUC-Y4F22D20**

Corning® ALTOS® HD Lite gel-free, single-jacket, single-armored cables with FastAccess® technology are designed for direct-buried installations.

The 24 fiber high-density buffer tube provides a 30 percent reduction in cable OD resulting in a 2x increase in fiber density. This improved density equals additional space for maximizing duct capacity. The innovative FastAccess technology feature combined with the gel-free loose tube design simplifies jacket removal and buffer tube access. The gel-free cable is fully waterblocked using craft-friendly water-swallowable materials for simple access with no clean up.

The loose tube design employs Corning's suite of optical fiber to provide reliable transmission parameters for a variety of voice, data, video, imaging and network applications. The flexible buffer tubes are easy to route in closures, and the SZ-stranded, loose tube design isolates fibers from installation and environmental rigors while allowing easy midspan access. The single-armored construction provides additional crush and rodent protection. These cables have a polyethylene jacket that is rugged, durable and easy to strip.

## Features and Benefits

### ALTOS® HD FastAccess® Technology

With the combination of a jacket with an innovative technology used to bind cable construction through the manufacturing process, eliminating the use of binder yarns and waterblocking tapes and up to a 60 percent improvement in cable access time. These technologies also reduce the overall risk of inadvertent fiber damage by reducing the need for sharp cable access tools.

### Polyethylene jacket

Rugged, durable and easy to strip (while providing superior protection against UV radiation, fungus, abrasion and other environmental factors)

### Fully waterblocked loose tube all-dielectric gel-free design

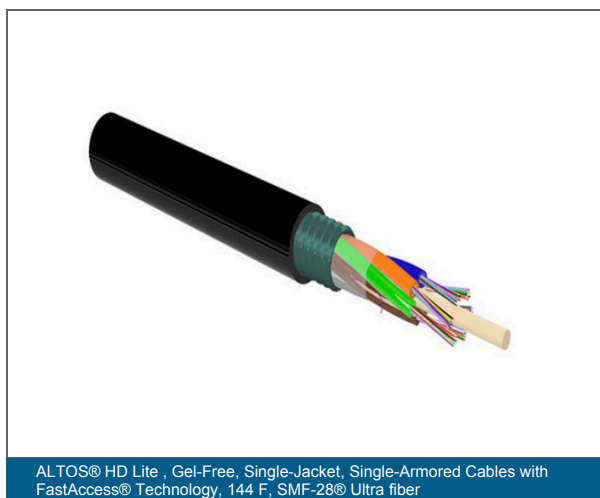
Simple access and no clean up

### Single-armored construction

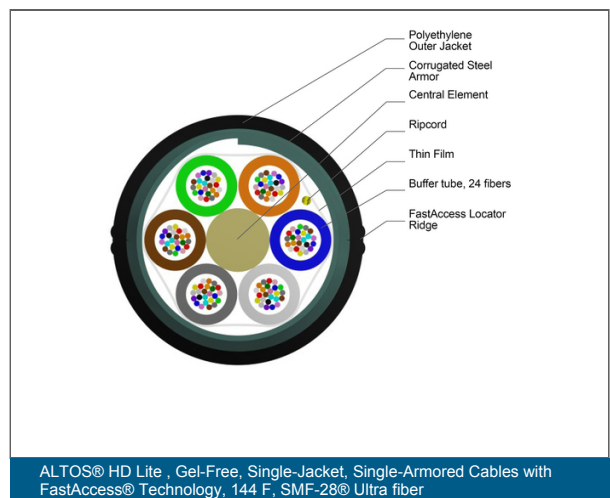
Provides additional crush and rodent protection

### Available in Corning's 62.5 µm and 50 µm multimode, standard single-mode (G.652.D, G.657.A1), dispersion shifted single-mode (G.655, G.654), and hybrid versions

Ready for any application from Gigabit Ethernet all the way to ≥ 800G Long Haul



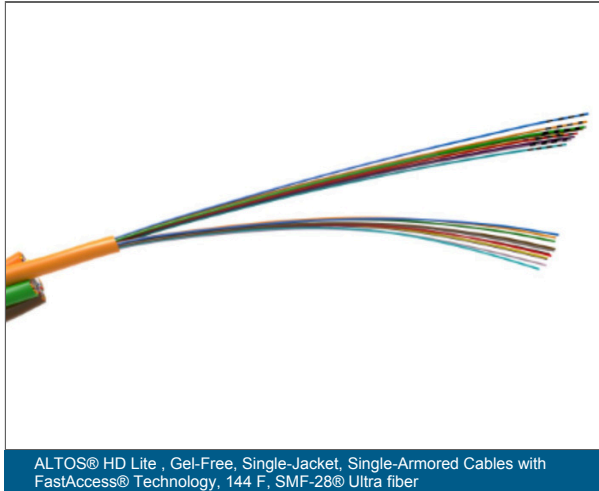
ALTOS® HD Lite , Gel-Free, Single-Jacket, Single-Armored Cables with FastAccess® Technology, 144 F, SMF-28® Ultra fiber



ALTOS® HD Lite , Gel-Free, Single-Jacket, Single-Armored Cables with FastAccess® Technology, 144 F, SMF-28® Ultra fiber

# ALTOS® HD Lite, Gel-Free, Single-Jacket, Single-Armored Cable with FastAccess® Technology 144F, SMF-28® Ultra fiber, Single-mode (OS2)

CORNING



ALTOS® HD Lite , Gel-Free, Single-Jacket, Single-Armored Cables with FastAccess® Technology, 144 F, SMF-28® Ultra fiber

## Specifications

### General Specifications

|                      |                             |
|----------------------|-----------------------------|
| Installation Methods | Aerial, Direct Buried, Duct |
| Cable Type           | Loose Tube                  |
| Environment          | Outdoor                     |
| Product Type         | Steel Tape Armor            |
| Fiber Category       | SMF-28® Ultra               |
| Cable geometry       | Round                       |

### Standards

|                          |                                                                                                                                |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| RoHS                     | Free of hazardous substances according to RoHS 2011/65/EU                                                                      |
| Common Installations     | Outdoor lashed aerial, duct and direct-buried, indoor when installed according to National Electrical Code® (NEC®) Article 770 |
| Design and Test Criteria | ANSI/ICEA S-87-640                                                                                                             |

# ALTOS® HD Lite, Gel-Free, Single-Jacket, Single-Armored Cable with FastAccess® Technology 144F, SMF-28® Ultra fiber, Single-mode (OS2)

CORNING

## Environmental Conditions

|                                 |                                                                                                                                                                                                                |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Notes                           | Corning recommends storing cable in a proper temperature environment prior to installation to allow the cable temperature to meet installation temperature range specifications for best installation results. |
| Temperature Range, Installation | -30 °C to 70 °C (-22 °F to 158 °F )                                                                                                                                                                            |
| Temperature Range, Operation    | -40 °C to 70 °C (-40 °F to 158 °F )                                                                                                                                                                            |
| Temperature Range, Storage      | -40 °C to 70 °C (-40 °F to 158 °F )                                                                                                                                                                            |

## Cable Design

|                          |                                                                                  |
|--------------------------|----------------------------------------------------------------------------------|
| Central Element          | Dielectric                                                                       |
| Fiber Count              | 144                                                                              |
| Outer Jacket Color       | Black                                                                            |
| Outer Jacket Material    | Polyethylene (PE)                                                                |
| Buffer Tube Color        | Blue, Orange, Green, Brown, Slate, White                                         |
| Buffer Tube Diameter     | 2.7 mm (0.11 in)                                                                 |
| Number of Active Tubes   | 6                                                                                |
| Number of Tube Positions | 6                                                                                |
| Tape                     | Thin Film                                                                        |
| Fiber Coloring           | Blue, Orange, Green, Brown, Slate, White, Red, Black, Yellow, Violet, Rose, Aqua |
| Fibers per Tube          | 24                                                                               |
| Color Code Standards     | Telcordia                                                                        |

## Mechanical Specifications

|                                   |                     |
|-----------------------------------|---------------------|
| Max. Tensile Strength, Long-Term  | 890 N (200.08 lbf)  |
| Max. Tensile Strength, Short-Term | 2700 N (606.98 lbf) |
| Nominal Outer Diameter            | 12.5 mm (0.49 in )  |
| Min. Bend Diameter Installation   | 376 mm (14.8 in)    |
| Min. Bend Diameter Operation      | 300 mm (11.81 in)   |

# ALTOS® HD Lite, Gel-Free, Single-Jacket, Single-Armored Cable with FastAccess® Technology 144F, SMF-28® Ultra fiber, Single-mode (OS2)

CORNING

## Optical Characteristics

|                         |                                      |
|-------------------------|--------------------------------------|
| Fiber Code              | Z                                    |
| Performance Option Code | 22                                   |
| Fiber Category          | OS2                                  |
| Fiber Type              | Single-mode (OS2) / 250 µm           |
| Fiber Name              | Bend-Improved Single-mode (OS2)      |
| Maximum Attenuation     | 0.34 dB/km / 0.34 dB/km / 0.22 dB/km |
| Wavelengths             | 1310 nm / 1383 nm / 1550 nm          |
| Fiber Compliance        | ITU-T G.652.D and ITU-T G.657.A1     |

## Dimensions

|              |                             |
|--------------|-----------------------------|
| Cable Weight | 139 kg/km (93.4 lb/1000 ft) |
|--------------|-----------------------------|



Corning Optical Communications LLC • 4200 Corning Place • Charlotte, NC • 28216 • United States  
800-743-2675 • FAX: 828-325-5060 • International: +1-828-901-5000 • [www.corning.com/opcomm](http://www.corning.com/opcomm)

A complete listing of the trademarks of Corning Optical Communications is available at [www.corning.com/opcomm/trademarks](http://www.corning.com/opcomm/trademarks). All other trademarks are the properties of their respective owners. Corning Optical Communications is ISO 9001 certified. © 2026 Corning Optical Communications. All rights reserved.